



**Faculty of Engineering & Technology Electrical & Computer Engineering
Department**

ENEE4113 - COMMUNICATION LAB

Experiment #6 PRELAB

Pulse Amplitude Modulation (Sampling)

Prepared By:

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Instructor: Dr.Alhareth Zayoud

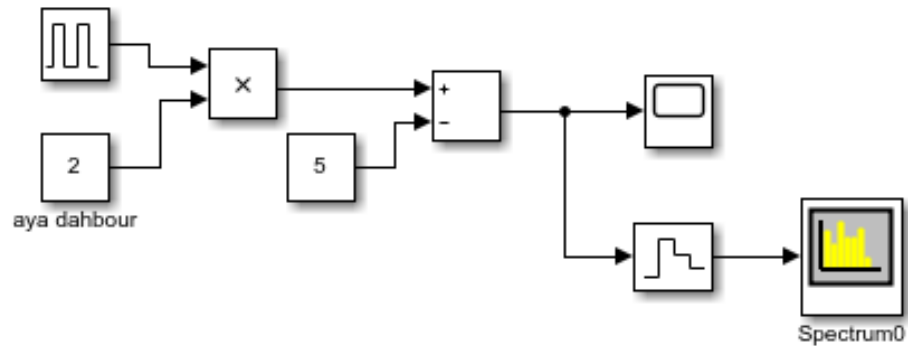
Assistant: Eng.Shadi

Section: 4

Date: 15/5/2023

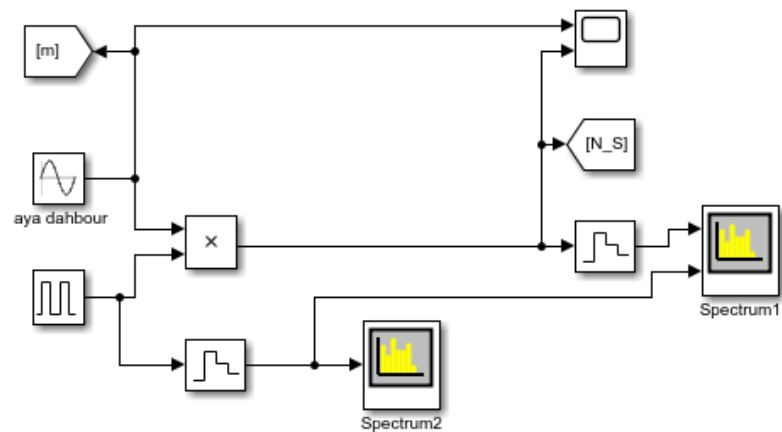
Part 1: Block Diagram Time and Frequency Characteristics of pulse train

- Time and Frequency Characteristics of pulse train:

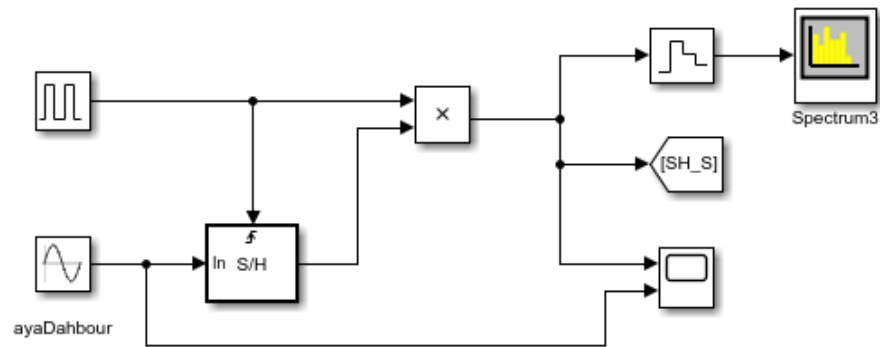


- Characteristics of Pulse Amplitude Modulation, Demodulation and aliasing (PAM):

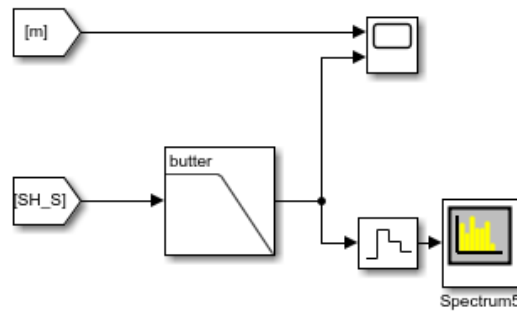
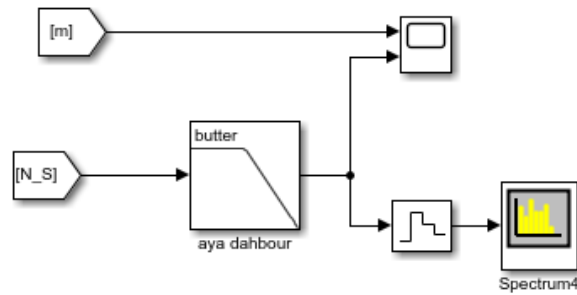
- Natural Sampling



- Sample and Hold (flat topped) Sampling



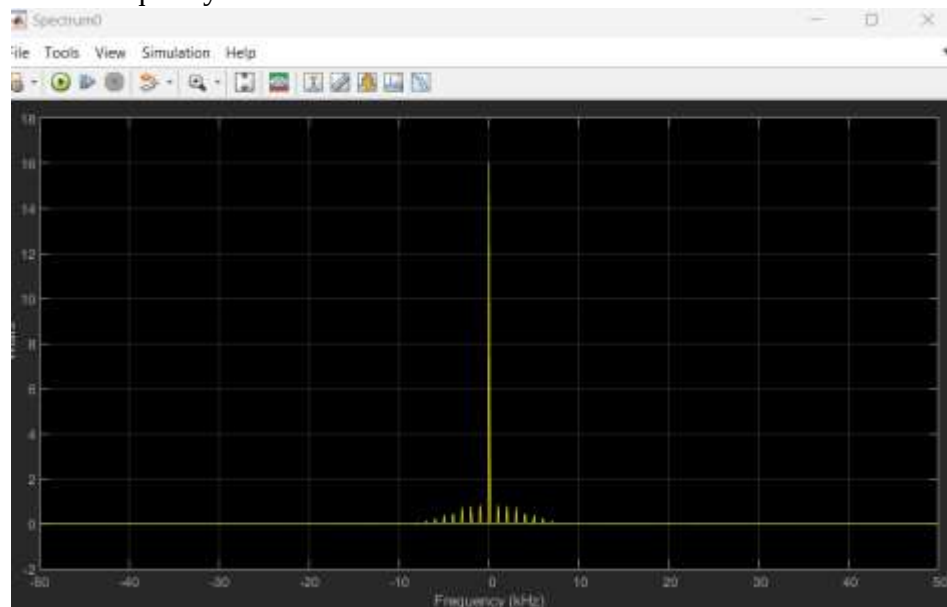
- Demodulation Natural Sampling



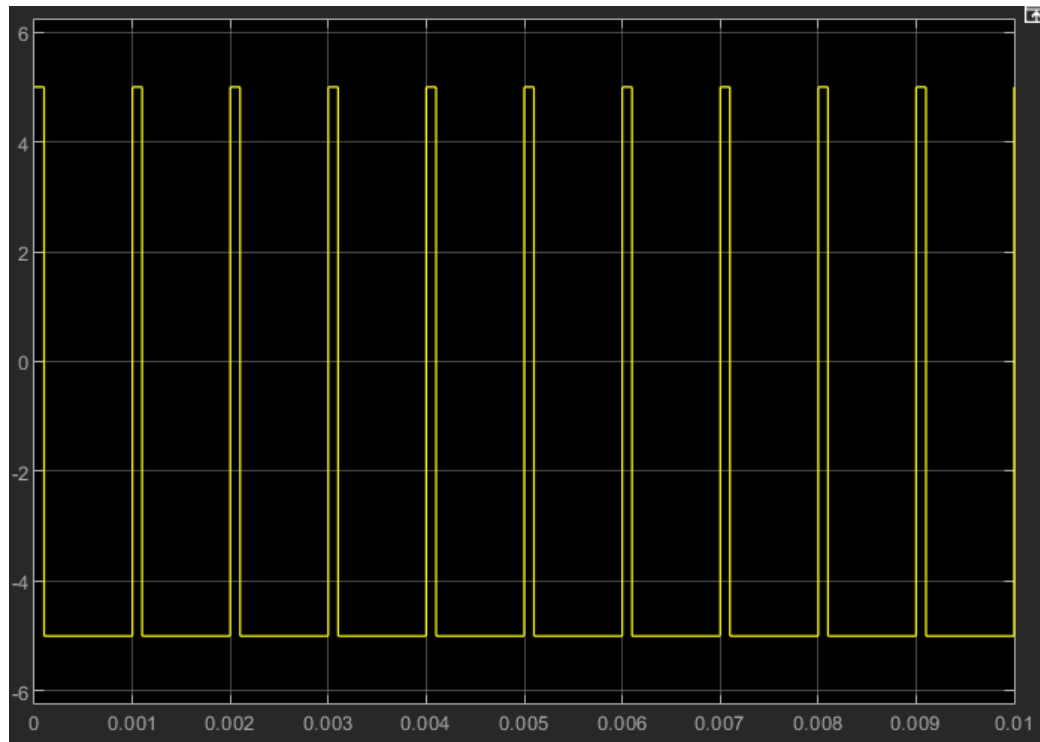
Part 2: The Results

➤ Pulse train

- In frequency domain



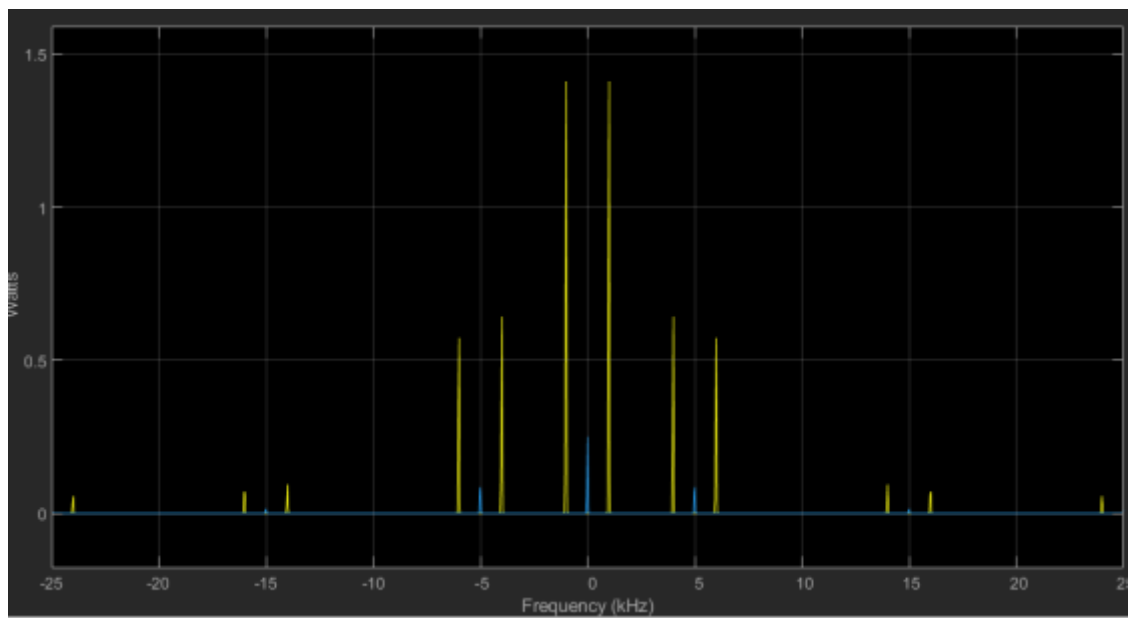
- In time domain



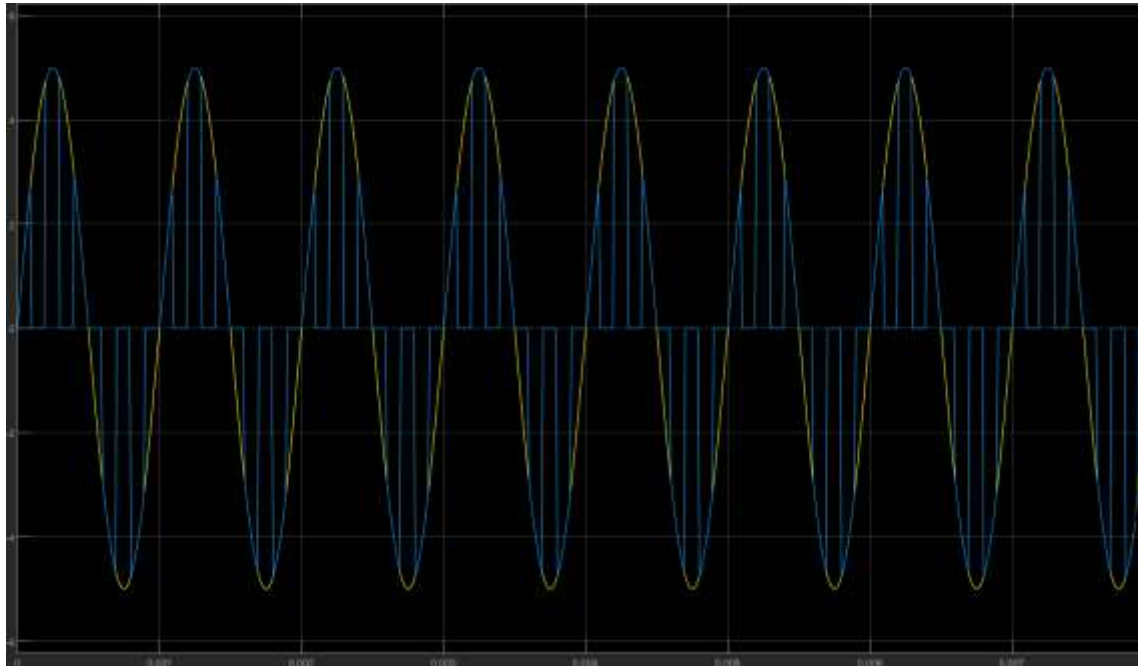
➤ **PAM**

1. At 1000kHz & duty cycle 50%

- In frequency domain

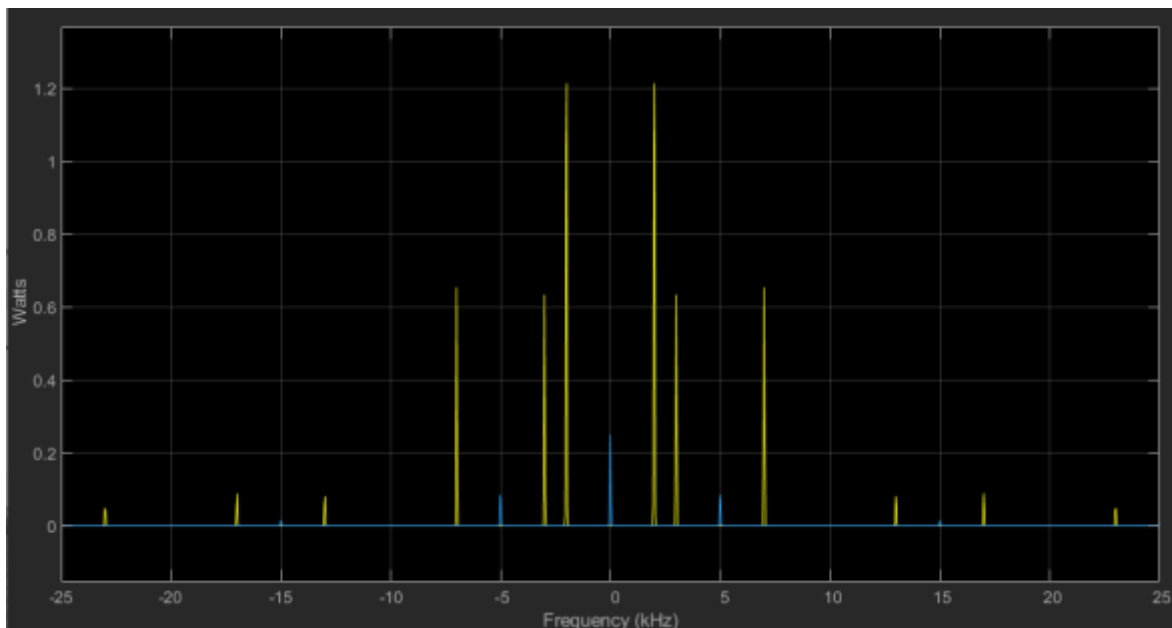


- In time domain

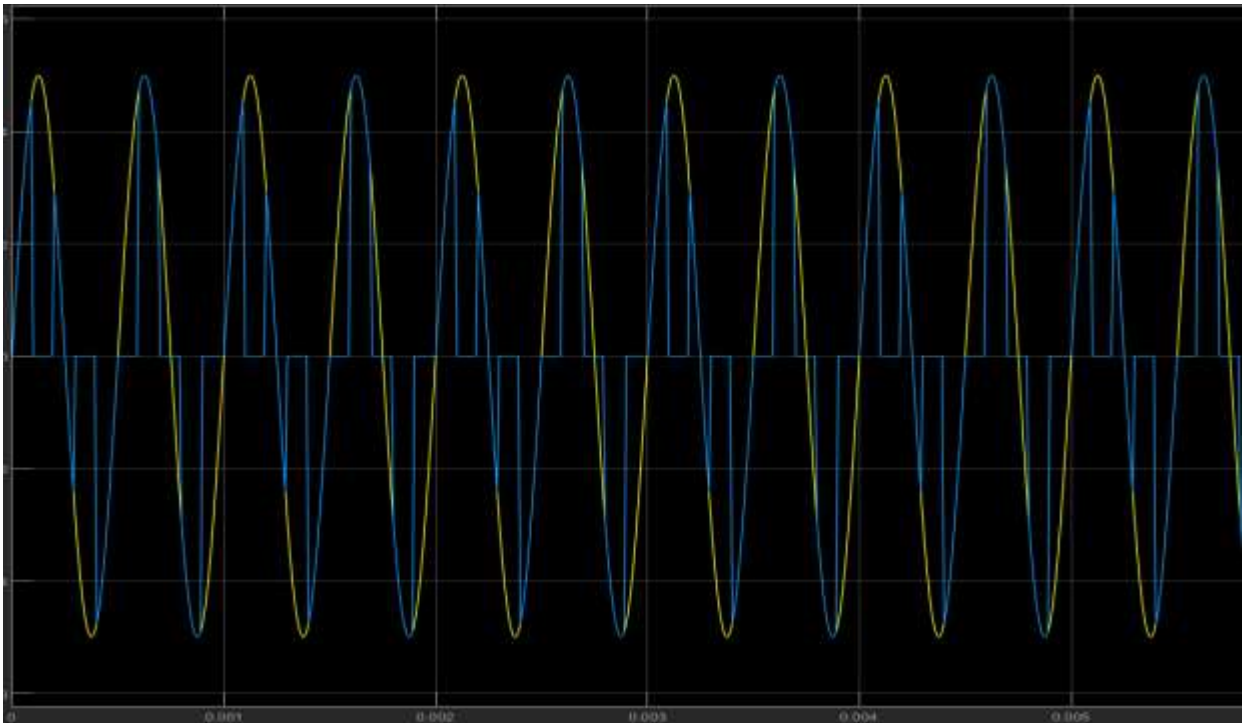


2. At 2000kHz & Pulse Width 50

- Frequency domain

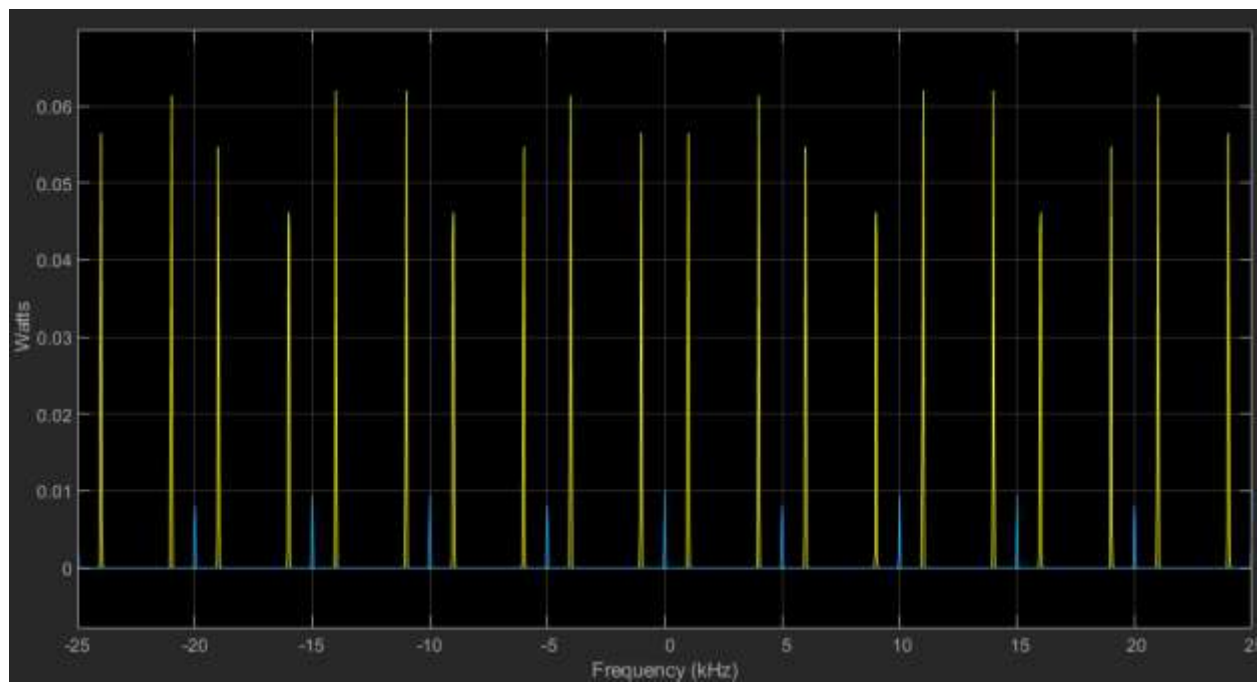


- Time domain

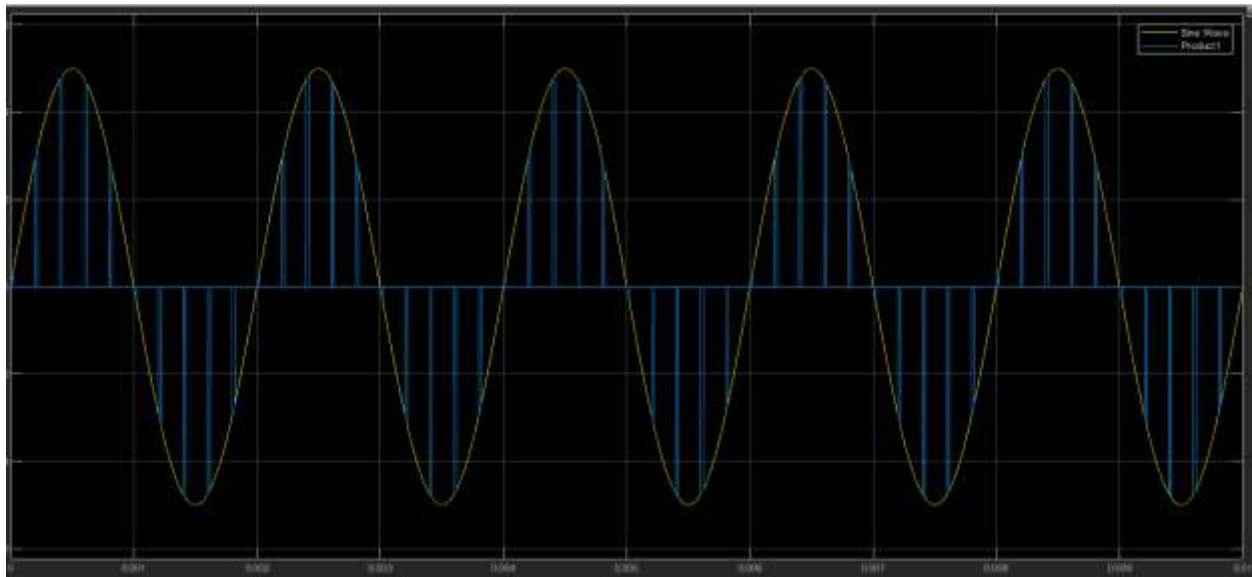


3. At 1000kHz & duty cycle 10%

- Frequency domain

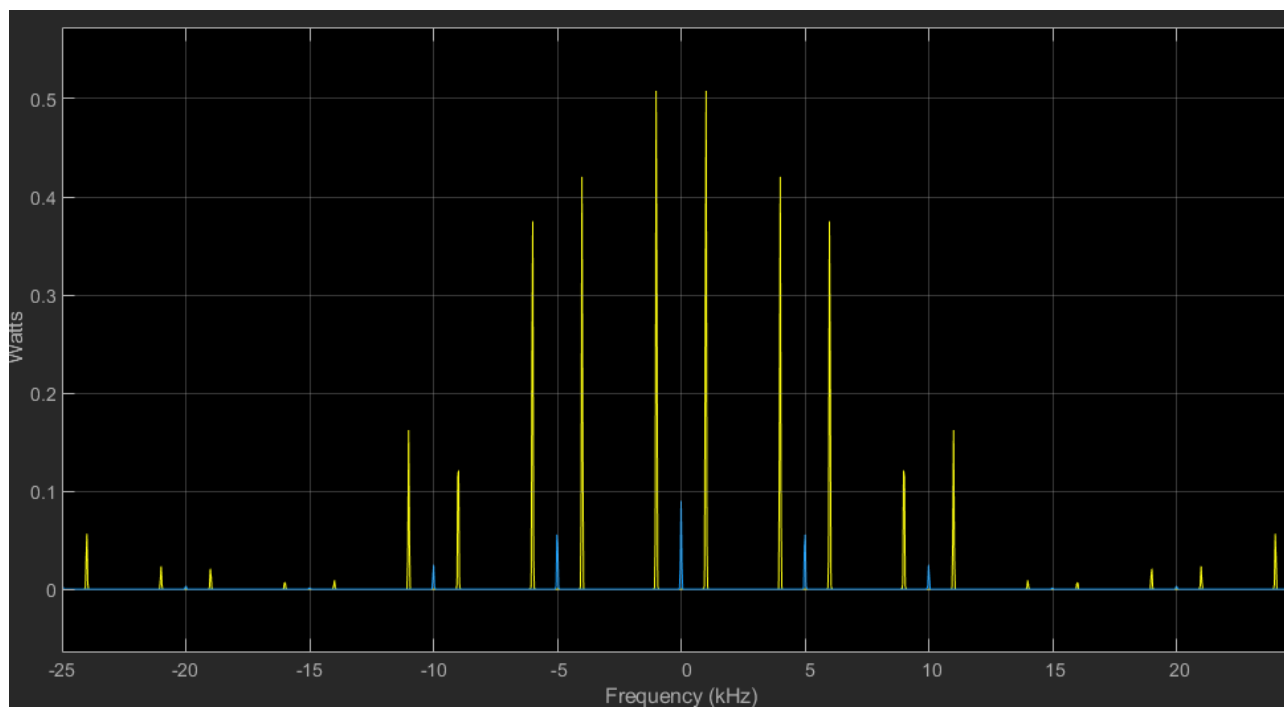


- Time domain

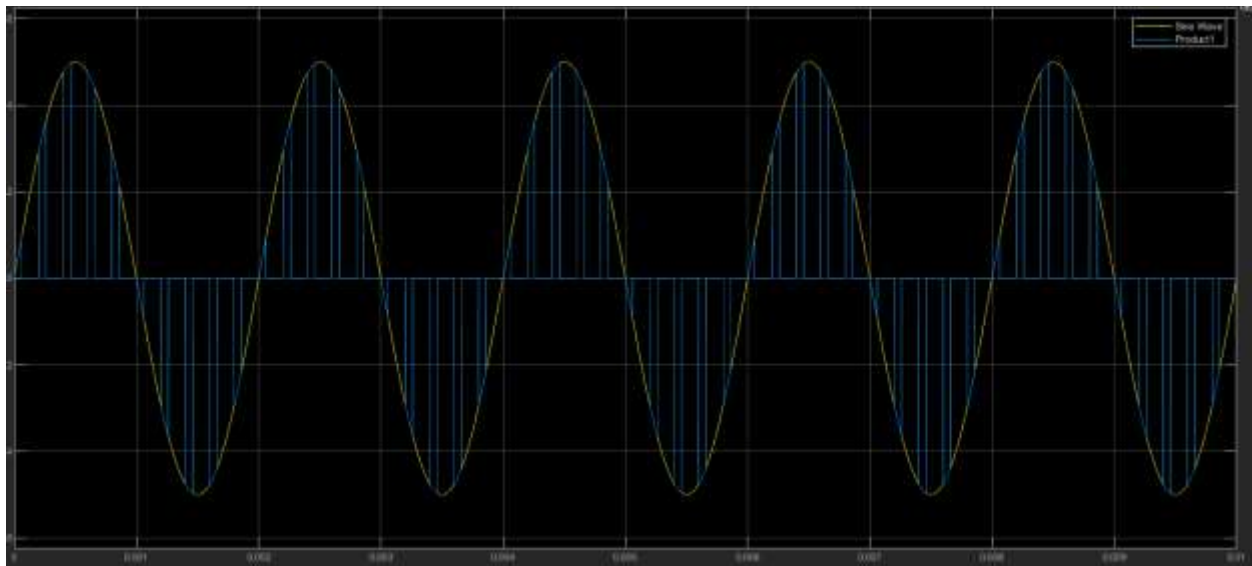


4. At 1000kHz & Pulse Width 30

- Frequency domain

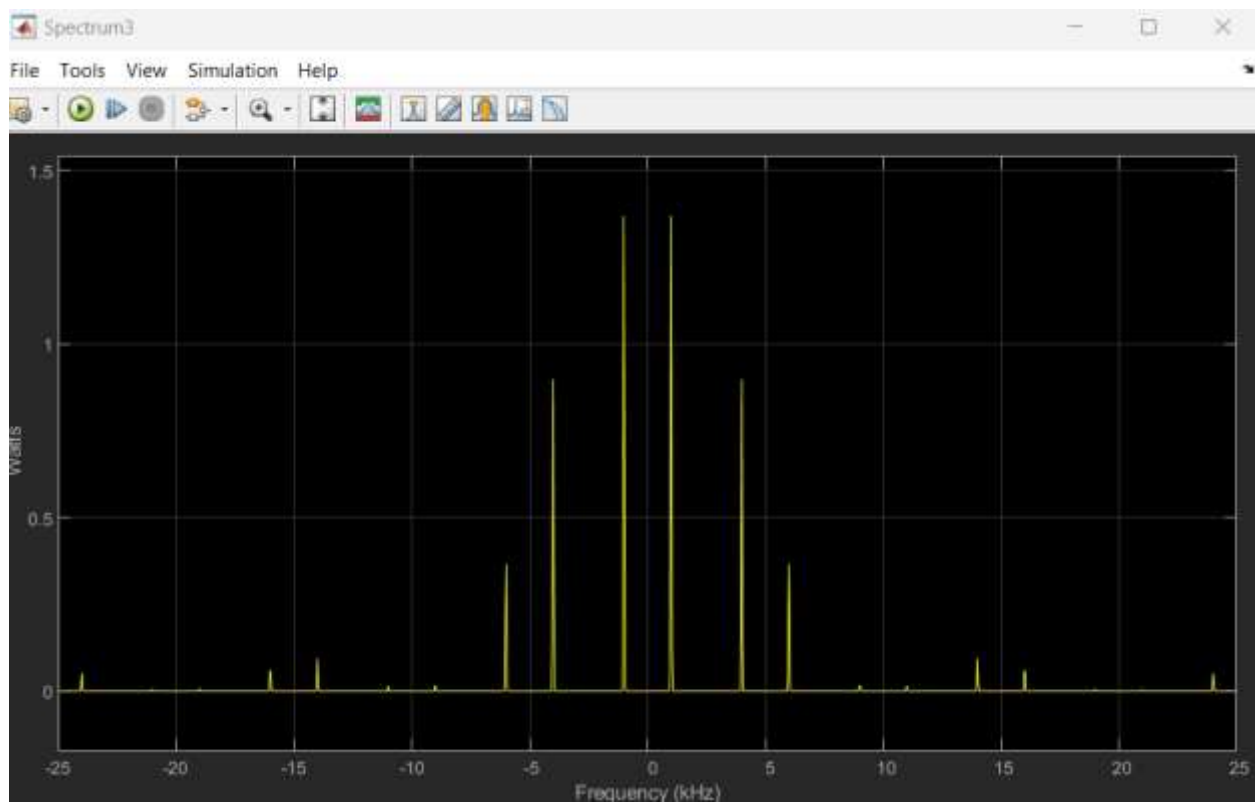


- Time domain

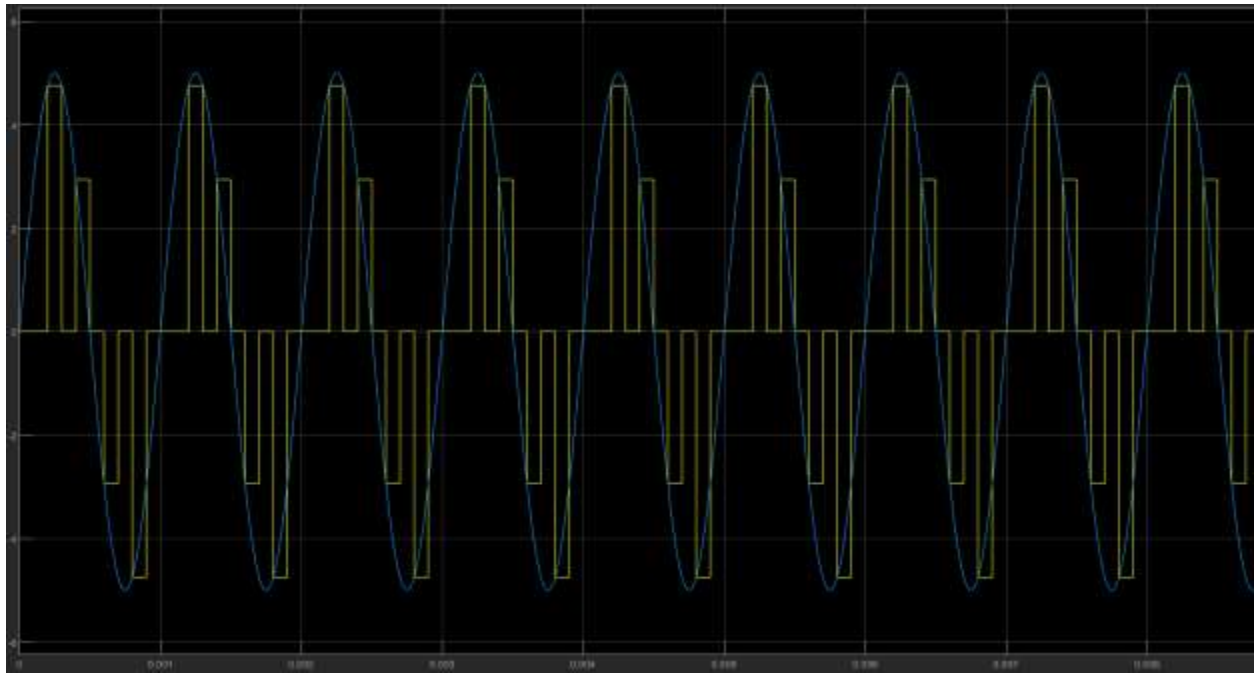


➤ **TDM**

- Frequency domain

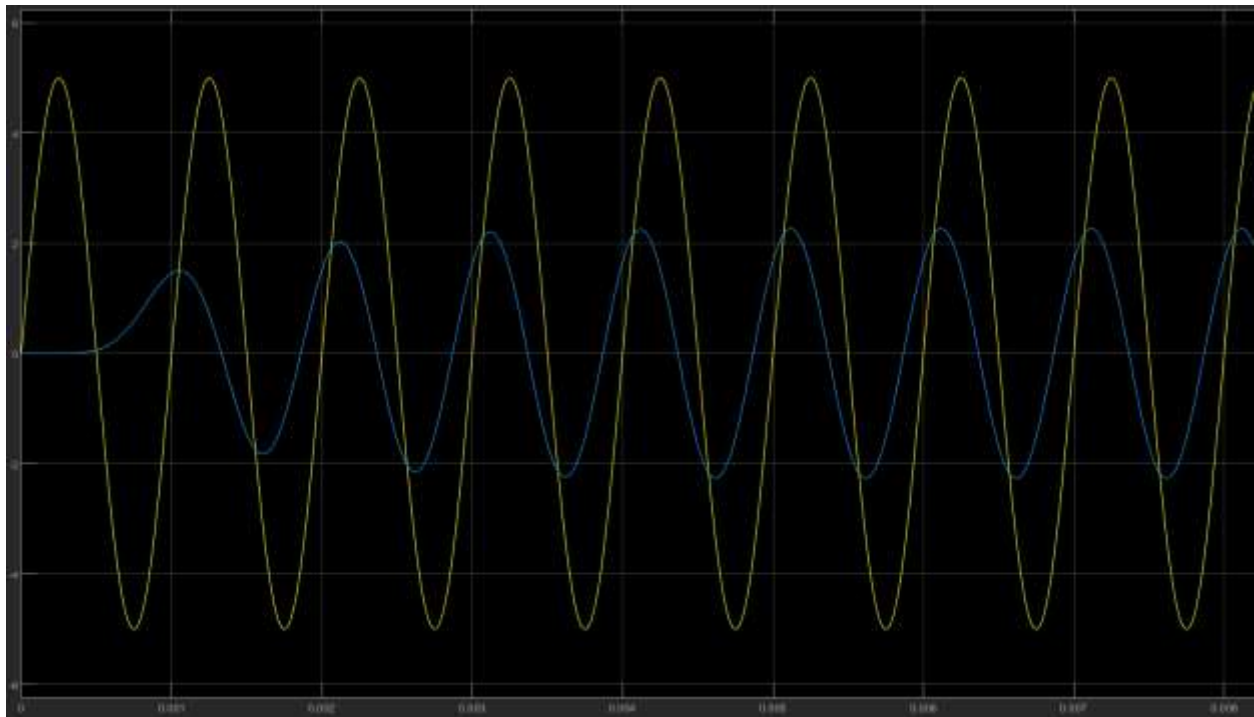


- Time domain

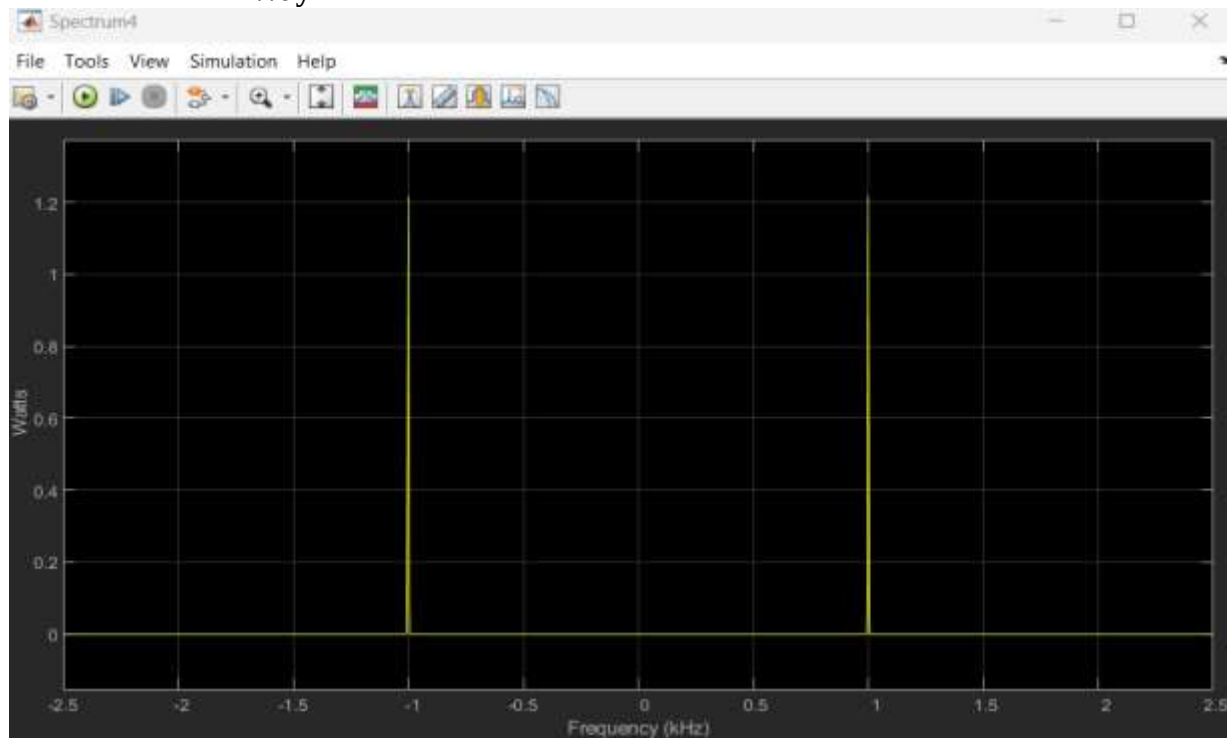


➤ **Demodulation**

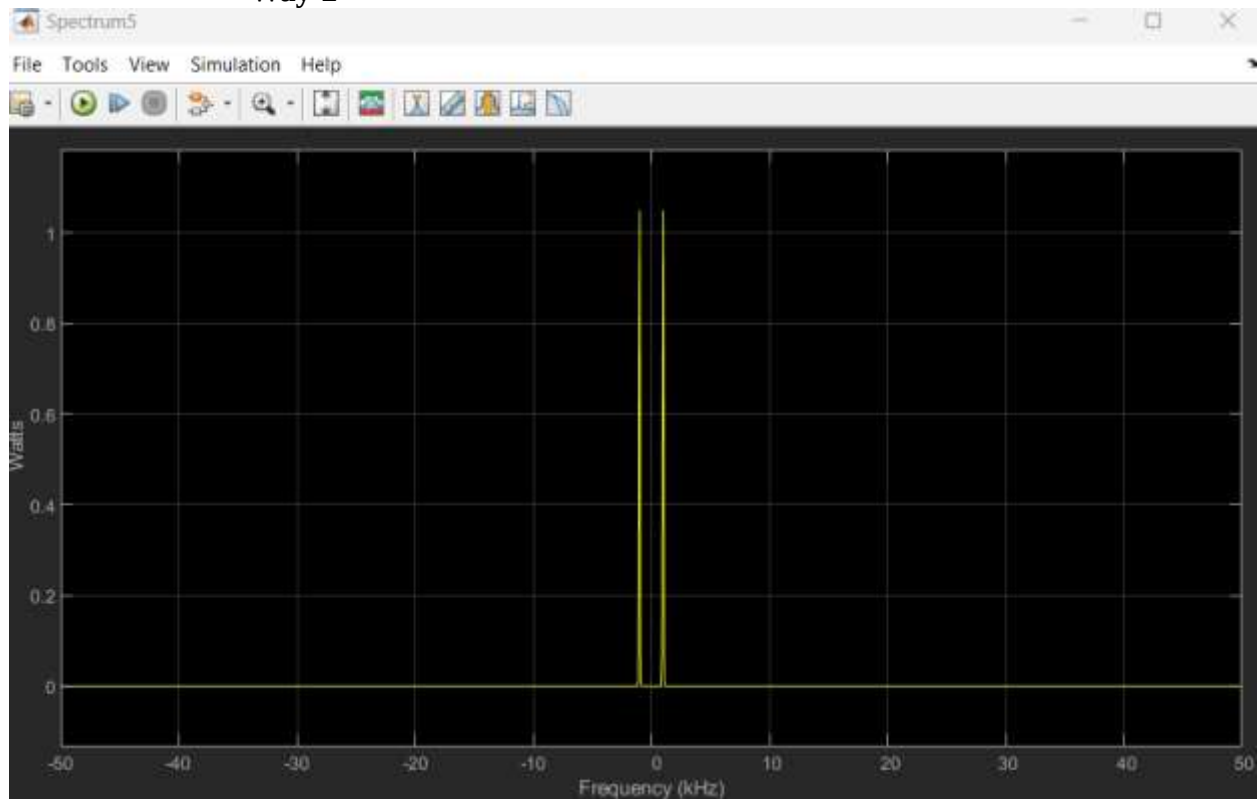
- Time domain



- Frequency domain
 - **Way 1**



- **Way 2**



**** Sitting:**

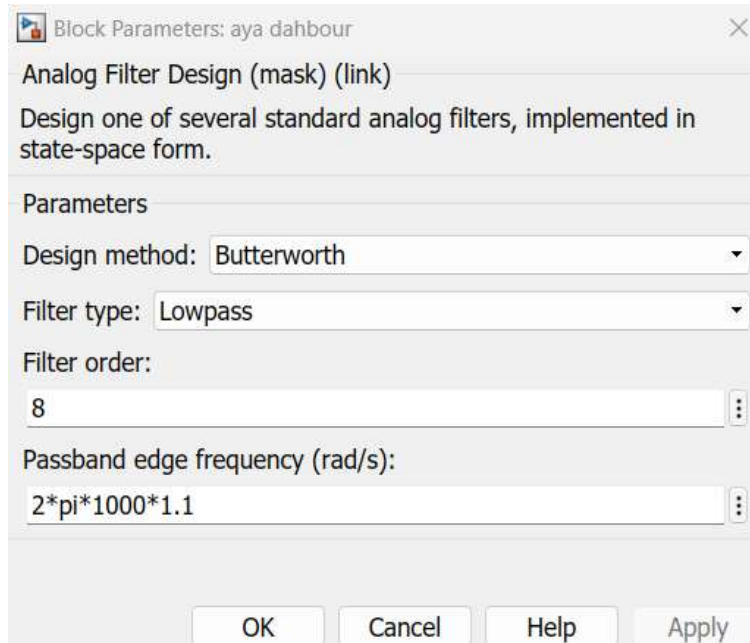


Figure 1: Filter sitting

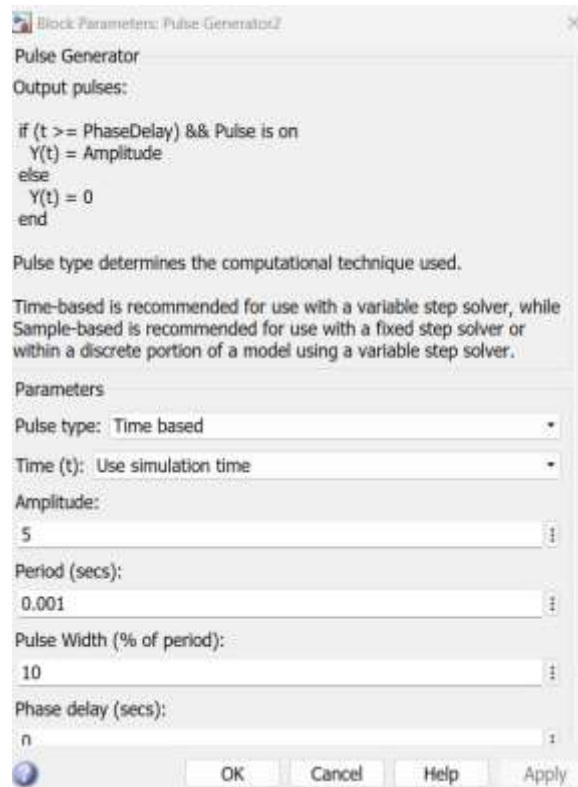


Figure 2: Pulse generator sitting

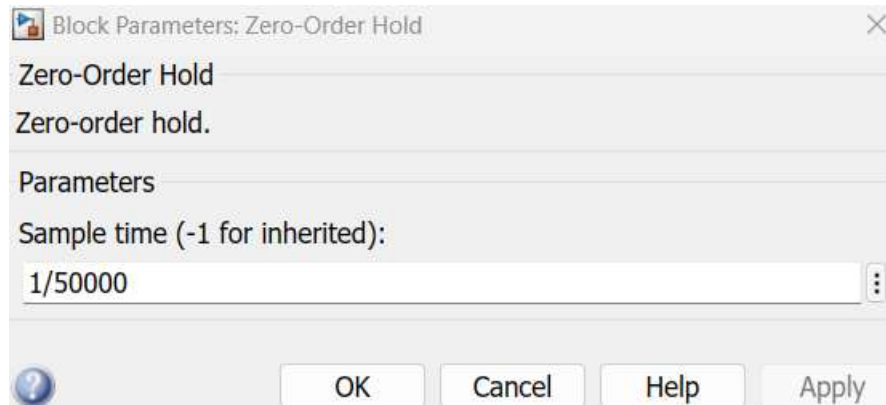


Figure 3: Zero Order Hold

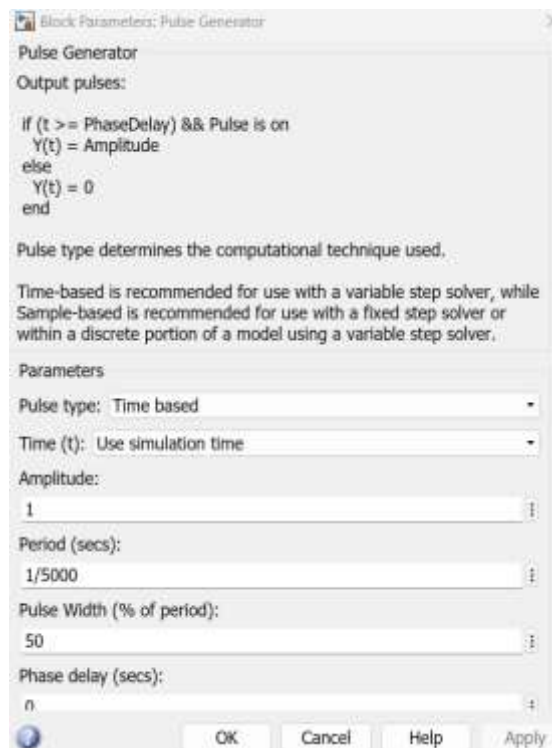


Figure 4: Pulse Generator 2

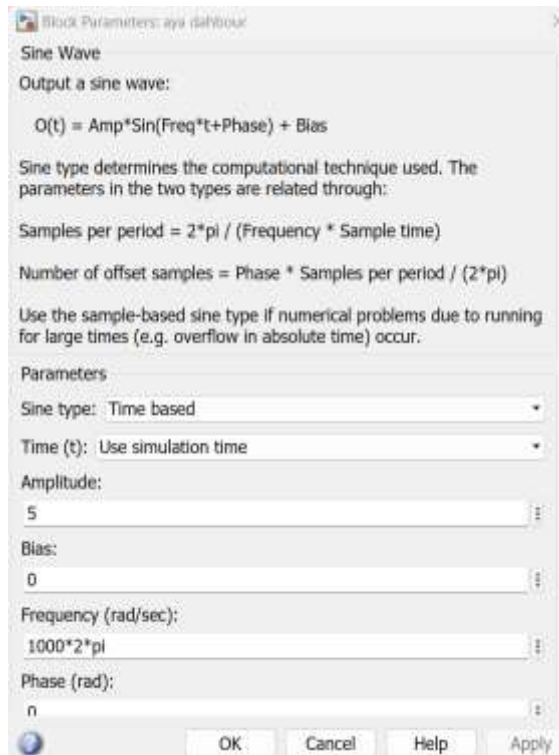


Figure 5: sine wave setting